

Abhishek Dhanani

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EDUCATION

Sacred Heart University (Pursuing)

Master's in computer science – data science track

Current GPA: 3.9

- Coursework: Machine learning, Deep Learning, Big Data Analysis

Fairfield, CT

Aug. 2024 – Dec. 2025

Government Engineering College

Bachelor's in computer science

- Coursework: Algorithms, Database Management, Advanced Mathematics, Software Engineering

Gandhinagar, Gujarat, India

Aug. 2020 – May 2023

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java

Data Visualization: Tableau, Matplotlib, Seaborn

Developer Tools: Git, Visual Studio, PyCharm, Jupyter notebook

Machine Learning Frameworks: Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Hugging Face Transformers

EXPERIENCE

Data analysis intern

Infolabz Pvt. Ltd.

Jan 2023 – July 2023

Ahmedabad, INDIA

- Cleaned and analyzed data using Pandas and NumPy to identify trends, patterns and actionable insights using tools like excel and Python.
- Developed visualizations and reports with Tableau and Matplotlib to present findings to teams.
- Contributed to the development of regression and classification models with Scikit-learn, to gain meaningful insights about data and analyze the market trends.
- Worked closely with marketing team to gather data requirements and deliver insights that supported strategic decision-making resulting in sales improvement.

PROJECTS

MailCast: Email to Audio Generation | NLP, Python, PyDub, OpenAI API, Gemini API

Mar 2025

- Developed a Python application to convert emails into audio podcasts by integrating Gmail API for email retrieval and BeautifulSoup for text preprocessing.
- Utilized a large language model (LLM) to generate podcast scripts from email content and OpenAI's Text-to-Speech (TTS) API with PyDub for audio output.
- Implemented OAuth2 authentication and chunked text processing to handle API token limits, ensuring scalability for large email datasets.
- Demonstrated skills in API integration, text processing, natural language generation, and audio generation.

RAG System for Document Summarization | Python, LangChain, ChromaDB, OpenAI Embedding, GPT models

Mar 2025

- Designed a Retrieval-Augmented Generation (RAG) system in Python to summarize lengthy legal documents, preprocessing text and storing summaries in a Chroma vector database.
- Integrated OpenAI embeddings and GPT models with LangChain to retrieve and generate concise, accurate summaries from document sections.
- Built a multi-vector retriever pipeline to combine original text and summaries, enhancing retrieval efficiency and summary coherence.
- Showcased expertise in NLP, vector databases, retrieval techniques, and large language models.

Fake News Classification using DistilBERT | NLP, Python, Hugging Face Transformers

Spring 2025

- Developed an NLP model to classify news articles as real or fake by fine-tuning DistilBERT using Hugging Face's Transformers library and AutoTokenizer for efficient text preprocessing.
- Processed a dataset of over 44,000 articles (Fake.csv and True.csv), performing data cleaning, tokenization, and splitting (70% train, 15% validation, 15% test).
- Fine-tuned DistilBERT on PyTorch with GPU, leveraging mixed precision training (FP16) for faster computation and reduced memory usage. Achieving an accuracy of 99.98% on the test set, with a confusion matrix and classification report for evaluation.
- Demonstrated skills in NLP, deep learning, model optimization, and Hugging Face ecosystem integration.